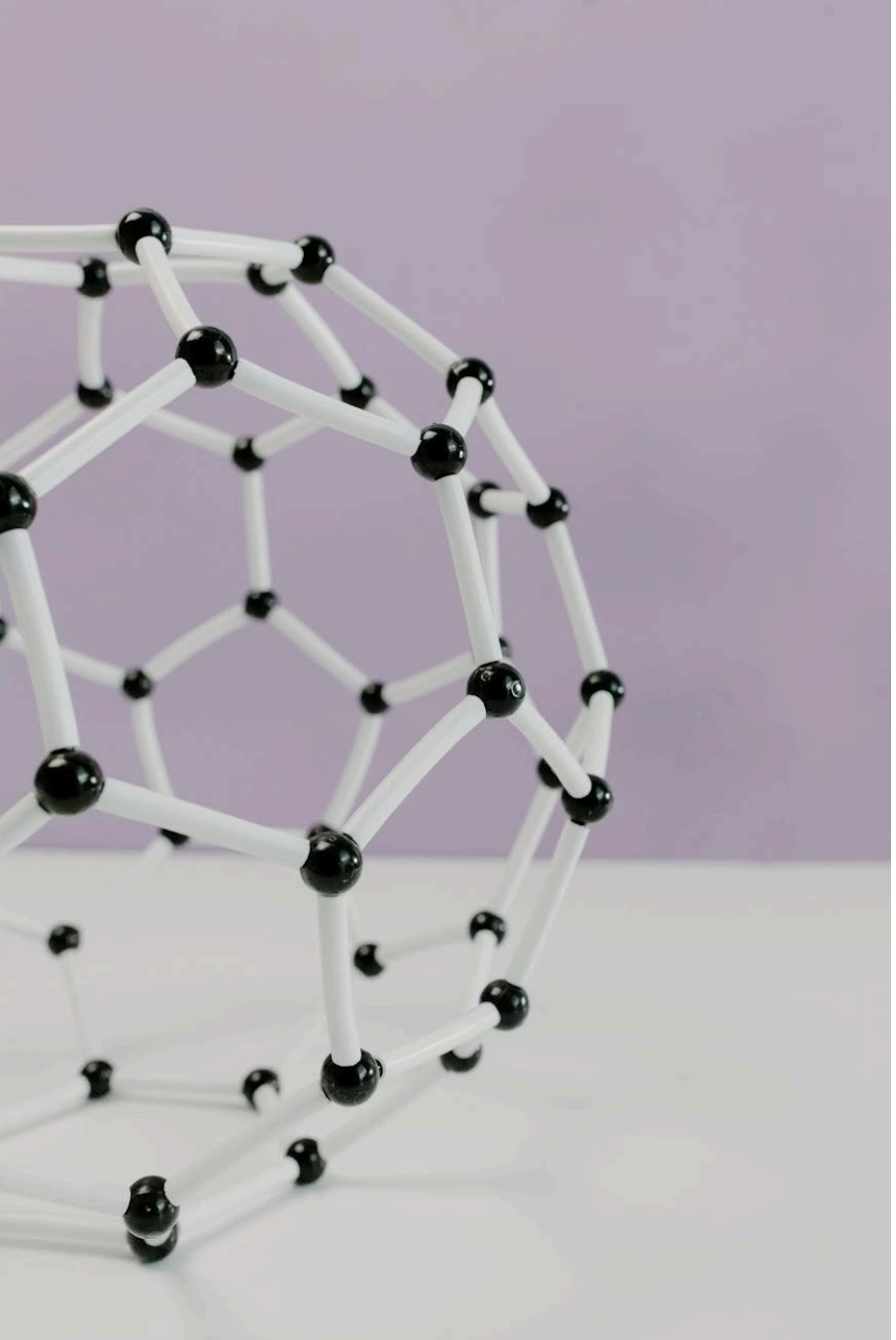


AP BIOLOGY · UNIT 1

TOPIC 1.2

Elements of Life

How atoms and molecules from the environment become the building blocks of every living organism — from the simplest bacterium to the most complex multicellular life form.

CHAPTER 1

Learning Objectives

Before we dive into the chemistry of life, let's establish what you need to master for this topic. These learning objectives align directly with the AP Biology Curriculum Framework and will guide our study throughout these slides.

1

Macromolecule Composition

Describe the composition of macromolecules required by living organisms. You should be able to identify which elements are present in each of the four major biological macromolecules — carbohydrates, proteins, lipids, and nucleic acids — and explain why these elements are essential.

2

Water's Unique Properties

Explain how the properties of water that result from its **polarity** and **hydrogen bonding** affect its biological function. Water is the universal solvent of life, and understanding its chemistry is foundational to understanding biochemistry.

3

Essential Knowledge Integration

Connect the roles of carbon, hydrogen, oxygen, nitrogen, sulfur, and phosphorus to specific macromolecules. Understand that atoms and molecules from the environment are necessary to build new molecules within living systems.

Building Blocks from the Environment

Why the Environment Matters

Living organisms are not closed systems. Every molecule in your body was once part of the non-living environment — the air you breathe, the water you drink, and the food you eat. This is a core concept in AP Biology:

> **Atoms and molecules from the environment are necessary to build new molecules.**

Organisms obtain raw materials (elements and simple compounds) from their surroundings and reassemble them into the complex biological molecules needed for structure, function, energy storage, and information transfer. This process is part of the broader concept of **matter cycling** through ecosystems — atoms are recycled, not created or destroyed.

The "Big Six" Elements of Life

While the periodic table contains over 100 elements, life primarily relies on just six. These elements make up approximately **99%** of the mass of most living organisms:

- **Carbon (C)** — the backbone of all organic molecules
- **Hydrogen (H)** — found in virtually every biological molecule
- **Oxygen (O)** — critical for energy metabolism and water
- **Nitrogen (N)** — essential for amino acids and nucleic acids
- **Phosphorus (P)** — key for energy transfer (ATP) and DNA
- **Sulfur (S)** — important for protein structure

Memory aid: **CHNOPS** — pronounce it "schnops" — is the classic mnemonic for remembering these six elements.

CHNOPS: The Six Elements of Life

Each of these six elements has unique chemical properties that make it indispensable to biological systems. Let's examine what makes each one special and where you'll find it in living organisms.

Carbon (C)

Carbon can form **four covalent bonds**, making it the most versatile building-block element. It creates the structural backbone (carbon skeleton) of all organic molecules — carbohydrates, lipids, proteins, and nucleic acids. Its ability to bond with itself in long chains, branched structures, and rings gives rise to the enormous diversity of organic compounds.

Hydrogen (H)

The simplest and most abundant element in the universe. In biology, hydrogen atoms are attached to carbon skeletons, participate in **hydrogen bonding** (critical for water's properties and protein folding), and play a central role in **energy transfer** through redox reactions and the electron transport chain.

Oxygen (O)

Highly electronegative, oxygen pulls electrons toward itself in covalent bonds, creating **polar molecules** like water. It is essential for aerobic cellular respiration, is a component of water (H_2O), and is found in many functional groups (hydroxyl, carboxyl, carbonyl, phosphate) that determine molecular behavior.

CHNOPS Continued: N, P, and S

While carbon, hydrogen, and oxygen form the bulk of biological molecules, nitrogen, phosphorus, and sulfur play specialized but absolutely critical roles. These elements often determine the *function* of a molecule, not just its structure.

🔵 Nitrogen (N)

Nitrogen is a key component of **amino acids** (the building blocks of proteins) and **nucleotides** (the building blocks of nucleic acids like DNA and RNA). Every amino acid contains an amino group ($-NH_2$), and the nitrogenous bases of nucleotides (adenine, guanine, cytosine, thymine, uracil) all contain nitrogen. Without nitrogen, there would be no genetic code and no enzymes.

🟡 Phosphorus (P)

Phosphorus is found in **phospholipids** (the molecules that form cell membranes) and **nucleic acids** (DNA and RNA). The sugar-phosphate backbone of DNA and RNA depends on phosphodiester bonds. Phosphorus is also central to **ATP** (adenosine triphosphate), the primary energy currency of cells. The phosphate groups in ATP store and release energy.

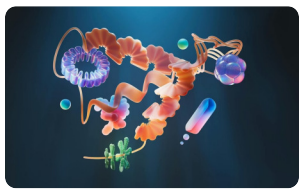
🟠 Sulfur (S)

Sulfur is specifically used in the building of **proteins**. Two amino acids — **cysteine** and **methionine** — contain sulfur. Cysteine is especially important because it can form **disulfide bonds** (S–S) with other cysteine residues, which are covalent cross-links that stabilize the three-dimensional (tertiary) structure of proteins.

AP Exam Tip: The College Board frequently tests whether you can match each element to its specific macromolecule(s). Remember: **N** → amino acids & nucleic acids; **P** → phospholipids & nucleic acids; **S** → proteins only.

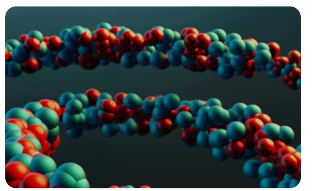
The Four Biological Macromolecules

Living organisms build four major classes of large, complex molecules — **macromolecules** — from simple subunits called monomers. The most prevalent elements in these molecules are **carbon, hydrogen, and oxygen**, but each class has its own elemental signature and biological role.



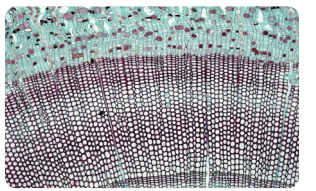
Carbohydrates

Elements: C, H, O
Monomers: Monosaccharides (glucose, fructose)
Functions: Quick energy source, structural support (cellulose in plant cell walls, chitin in arthropod exoskeletons), cell-to-cell recognition. Ratio of H:O is typically 2:1, reflecting the general formula $(CH_2O)_n$.



Proteins

Elements: C, H, O, N, **S**
Monomers: Amino acids (20 types)
Functions: Enzymes, structural support, transport, immune defense, signaling, movement. Sulfur appears in cysteine and methionine, enabling disulfide bridges that shape protein folding.



Lipids

Elements: C, H, O, **P** (in phospholipids)
Monomers: Glycerol + fatty acids
Functions: Long-term energy storage, cell membranes (phospholipid bilayer), insulation, hormones. Lipids are largely hydrophobic because of their long hydrocarbon chains.

Nucleic Acids

Elements: C, H, O, **N**, **P**
Monomers: Nucleotides
Functions: Store and transmit genetic information (DNA), direct protein synthesis (mRNA), energy transfer (ATP). The sugar-phosphate backbone uses phosphorus; nitrogenous bases use nitrogen.

Element-to-Macromolecule Map

One of the most testable skills on the AP exam is knowing which elements are found in which macromolecules. This table summarizes the relationships at a glance. Notice that **C, H, and O** are universal — they appear in all four classes — while **N, P, and S** are more selective.

Macromolecule	C	H	O	N	P	S
Carbohydrates	✅	✅	✅	—	—	—
Lipids	✅	✅	✅	—	✅*	—
Proteins	✅	✅	✅	✅	—	✅
Nucleic Acids	✅	✅	✅	✅	✅	—

*Phosphorus is present specifically in **phospholipids**, not in all lipids. Simple fats (triglycerides) contain only C, H, and O.

Key Takeaway: Carbon, hydrogen, and oxygen are the three most prevalent elements in biological molecules. Nitrogen is shared between proteins and nucleic acids. Phosphorus links phospholipids and nucleic acids. Sulfur is exclusive to proteins. Memorize this table — it's AP gold.

Water: The Molecule That Makes Life Possible

Water (H_2O) is arguably the most important molecule for life on Earth. Its unique physical and chemical properties — all arising from its **polarity** and ability to form **hydrogen bonds** — create the conditions necessary for biochemical reactions, temperature regulation, and molecular transport. Understanding water is so fundamental that the AP Biology exam dedicates an entire essential knowledge area to it.

Polarity Explained

Water is a **polar molecule** because oxygen is far more electronegative than hydrogen. Oxygen pulls the shared electrons closer to itself, creating a partial negative charge (δ−) near the oxygen and partial positive charges (δ+) near the two hydrogens. This unequal sharing of electrons produces a bent molecular geometry with a permanent dipole.

Hydrogen Bonding

Because of water's polarity, the δ+ hydrogen of one water molecule is attracted to the δ− oxygen of a neighboring water molecule. This intermolecular attraction is called a **hydrogen bond**. While individually weak (about 1/20 the strength of a covalent bond), hydrogen bonds are collectively powerful when millions of water molecules interact simultaneously.

Emergent Properties of Water

Water's polarity and hydrogen bonding give rise to several **emergent properties** — behaviors that arise from the collective interactions of many water molecules. Each of these properties has direct and significant biological consequences.

1

Cohesion & Adhesion

Cohesion is the attraction of water molecules to each other via hydrogen bonds, creating **surface tension** (insects can walk on water). **Adhesion** is water's attraction to other polar surfaces. Together, cohesion and adhesion drive **capillary action**, which helps water travel upward against gravity in plant xylem vessels — essential for transporting water and dissolved minerals from roots to leaves.

2

High Specific Heat

Water has a remarkably high **specific heat capacity** (4.18 J/g·°C) — it absorbs or releases a large amount of heat energy with only a small change in temperature. This is because hydrogen bonds must be broken before temperature rises. Biologically, this means organisms with high water content resist rapid temperature fluctuations, and large bodies of water **moderate coastal climates**.

3

High Heat of Vaporization

It takes a lot of energy to convert liquid water to gas because many hydrogen bonds must be broken. When water evaporates from a surface, it carries away significant thermal energy — this is **evaporative cooling**. Sweating in humans, transpiration in plants, and panting in dogs all exploit this property to prevent overheating.

4

Universal Solvent

Water's polarity makes it an excellent solvent for **polar and ionic substances** (hydrophilic molecules). It dissolves salts, sugars, amino acids, and many gases. This is critical because most biochemical reactions occur in **aqueous solution** — reactants must be dissolved to interact. Nonpolar molecules (hydrophobic) do not dissolve, which is why lipids form membranes.

5

Ice Floats (Density Anomaly)

Unlike most substances, water is **less dense as a solid** than as a liquid. When water freezes, hydrogen bonds lock molecules into an open crystalline lattice, making ice about 10% less dense. Ice floats and insulates the liquid water below, allowing **aquatic organisms to survive winter**. If ice sank, lakes and oceans would freeze from the bottom up, destroying aquatic ecosystems.

Water's Properties in Action

Let's see how these properties connect to real biological scenarios — the kind of connections the AP exam loves to test. Understanding these applications transforms memorization into genuine comprehension.

Transpiration in Trees

Water's **cohesion** (water sticking to water) and **adhesion** (water sticking to xylem cell walls) work together with evaporation at the leaf surface to pull continuous columns of water from roots to the tops of the tallest trees — sometimes over 100 meters high. This transpiration-cohesion-tension mechanism requires no metabolic energy from the plant; it's powered entirely by water's physical properties and solar energy.

Frozen Lake Ecosystems

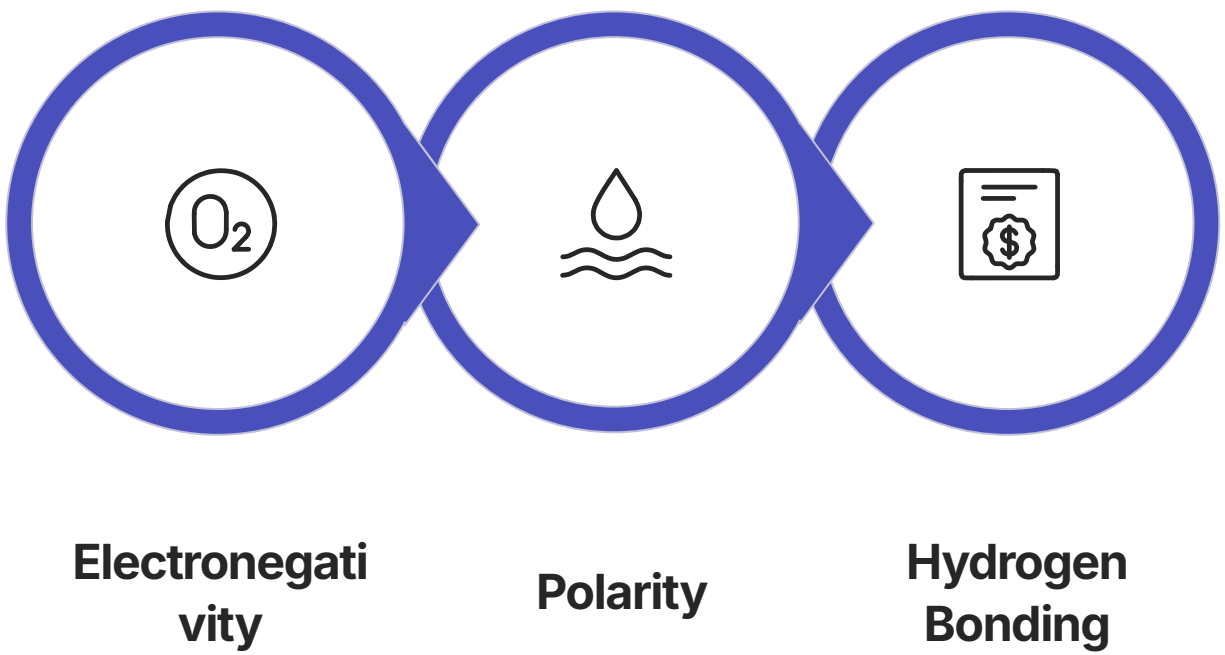
When winter arrives, surface water cools to 0°C and freezes. Because ice is less dense, it forms a floating insulating layer on top, while the liquid water below stays near **4°C** — the temperature at which water is most dense. This temperature gradient and insulation allow fish, amphibians, and other aquatic organisms to survive through even the harshest winters. Without this anomaly, Earth's freshwater ecosystems would collapse.

Evaporative Cooling

During intense exercise, your body temperature can rise dangerously. Sweat glands release water onto the skin surface, and as that water evaporates, it absorbs heat from the body (high heat of vaporization). This **evaporative cooling** is one of the most efficient thermoregulation mechanisms in mammals. Plants achieve something similar through **transpiration** — water evaporating from leaf stomata cools leaf surfaces.

Polarity, Hydrogen Bonding & Biological Function

Let's trace the logical chain that the AP exam expects you to understand. Every emergent property of water can be traced back to the same molecular cause:



This causal reasoning is exactly what free-response questions test. You should be able to explain: **Oxygen's high electronegativity → polar covalent bonds in H_2O → partial charges on the molecule → hydrogen bonds between water molecules → emergent properties (cohesion, adhesion, high specific heat, high heat of vaporization, versatile solvent, ice floats) → biological functions** (transport in plants, temperature regulation, aqueous biochemistry, aquatic habitat preservation). Practice tracing this chain from molecular cause to organismal effect.

REVIEW

Key Takeaways & Exam Prep

🧬 CHNOPS

Carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur are the six most important elements in biology. C, H, and O are in ALL four macromolecules. N is in proteins & nucleic acids. P is in phospholipids & nucleic acids. S is in proteins only.

🔬 Macromolecules

Know the four classes (carbs, lipids, proteins, nucleic acids), their monomers, their elemental composition, and their functions. Atoms and molecules from the environment are recycled and reassembled into these biological molecules.

💧 Water

Polarity → hydrogen bonding → emergent properties. Memorize all five properties (cohesion/adhesion, high specific heat, high heat of vaporization, universal solvent, ice floats) and connect each to a biological example.

Practice Free-Response Prompt

> *"Explain how water's molecular structure gives rise to TWO properties that are essential for the survival of organisms. For each property, describe the molecular basis and provide a specific biological example."*

Try answering this in a timed 10-minute window. Use the cause-and-effect chain from the previous slide: start at the molecular level (polarity, hydrogen bonds) and connect to the organism or ecosystem level. This type of multi-level reasoning earns full credit on the AP exam. Good luck! 🎯